

Characteristics of a Computer

1. Automatic

What does automatic mean?

Automatic means:

Once a computer is given the required data and instructions, it can perform the task automatically without continuous human intervention.

For example, suppose you write a program:

Input two numbers
Add them
Display the answer

You enter:

10
20

The computer automatically:

10 + 20
↓
30

You don't have to tell it:

"Now add them."

The program already contains that instruction.

Real-life example

An ATM is a good example.

You:

Insert card
↓
Enter PIN
↓

Select withdrawal



Enter amount

After you provide the required input, the system automatically performs the programmed operations.

Important limitation

"Automatic" does **not** mean the computer thinks independently like a human.

It means:

The computer automatically executes the instructions that humans/programmers have provided.

This distinction is very important.

2. Speed

Computers can perform operations extremely quickly.

A human might need considerable time to perform thousands or millions of calculations.

A computer can perform huge numbers of operations in a very short period.

For example, a computer can process:

$10 + 20$

50×100

$125 \div 5$

...

very rapidly.

Computer speed is often measured using:

- milliseconds
- microseconds
- nanoseconds

A **nanosecond** is:

0.000000001 second

That's one billionth of a second.

Important idea

Computer speed doesn't mean:

"The computer understands faster."

It means:

Its electronic circuits can execute operations extremely quickly.

3. Accuracy

Computers are capable of producing highly accurate results **when the data, program, and instructions are correct.**

Example:

$$1250 \times 25 = 31250$$

A computer can perform this calculation consistently.

But there is an important rule:

Wrong input or wrong instructions can produce a wrong result.

This is sometimes expressed as:

GIGO

Garbage In → Garbage Out

For example:

Correct data + Correct program



Correct result

But:

Wrong data



Processing



Wrong result

So don't say:

"Computers are always correct."

Instead say:

Computers can provide highly accurate results when the input and instructions are correct.

4. Versatility

Versatile means:

Capable of performing many different types of tasks.

The same computer can be used for:

Programming



Web browsing



Gaming



Video editing



Graphic design



Data analysis



Word processing



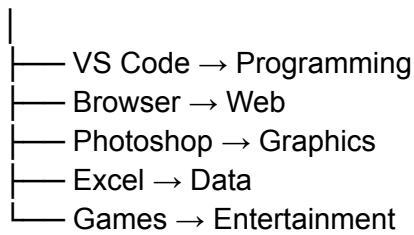
Scientific calculations

How can one machine do all these different things?

Because we can run **different programs** on it.

For example:

Computer



The hardware is the same, but different software gives it different capabilities.

Key idea:

Programmability is one major reason computers are versatile.

5. Reliability

Reliability means:

A computer can perform the same operation repeatedly and consistently without getting tired or changing its behavior due to boredom or emotions.

Suppose you ask a computer to perform:

1000×500

Then ask it 10,000 times.

If the same conditions and correct program are used, it will consistently perform the operation.

A computer doesn't say:

"I've calculated this too many times. I'm tired."

It doesn't get bored, angry, or distracted.

6. Diligence

Diligence means:

The ability to perform repetitive tasks continuously without fatigue, boredom, or loss of concentration.

Suppose a program has to process:

1,000,000 records

A human would naturally need:

Breaks

Rest

Sleep

A computer can continue processing as long as:

- it has power,
- the hardware works,
- and the program is running correctly.

Reliability vs Diligence

These two are easy to confuse.

Diligence:

Doesn't get tired or bored while repeatedly performing tasks.

Reliability:

Performs tasks consistently and dependably.

Think:

Diligence → "Doesn't get tired."

Reliability → "Works consistently."

7. Memory

Computers can store huge amounts of data.

For example:

Documents

Pictures

Videos

Programs

Music
Databases
Emails

Computer memory/storage allows information to be stored and retrieved when needed.

There are different types of memory/storage, which you'll study in detail later.

For now, understand:

Memory allows a computer to store data, instructions, and results for current or future use.

8. Programmable

We already discussed this in Lecture 1, but it is also a major characteristic.

A computer can be programmed to perform different tasks.

For example:

Program A → Calculate salary
Program B → Play music
Program C → Edit photos
Program D → Run a game
Program E → Analyze data

Therefore:

A computer is not limited to one specific operation. Its behavior can be changed by changing the program.

Characteristic	Simple meaning
Automatic	Executes instructions automatically
Speed	Performs operations very quickly
Accuracy	Produces precise results when input/program are correct
Versatile	Performs many different kinds of tasks

Reliable	Produces consistent results under the same conditions
Diligent	Doesn't get tired or bored from repetitive work
Memory	Stores data, instructions, and results
Programmable	Can be instructed to perform different tasks